STAT6207 • HW4 • EXPERIMENT REPORT

Keyword-Based CV–JD Matching

Local LLM keyword extraction · keyword clustering · autoencoder retrieval

10 GT pairs | 110 CVs | Top-10 retrieval

Data: Anonymised-CVs-100 + JD + Hired CV examples

Model: Qwen2.5-Coder-14B-Instruct (LM Studio, local)

Date: 2026-11-07

Abstract

This project studies CV–JD matching when both job descriptions (JDs) and CVs are unstructured natural language. The pipeline is: local-LLM keyword extraction → keyword merging/clustering → autoencoder encoding → vector retrieval. The data contain 10 standard JD–CV pairs with the actual hired CV, plus 100 anonymised CVs, giving a 110-CV candidate pool. Keywords are extracted by a local Qwen2.5-Coder-14B-Instruct model; clustering, encoding and retrieval are non-LLM programs. The primary evaluation is leave-one-pair-out (LOPO): each fold trains on 9 GT pairs + 100 unlabelled CVs and evaluates the held-out pair. The primary metric is Recall@10, with Recall@5, MRR@10, mean rank and Wilson 95% CI.

> **Key result:** LOPO Recall@10 = **0.800**; TF-IDF baseline = **0.800**; Δ = **+0.000**; in-sample upper bound = **1.000**.

1. Task and Data

1.1 Task definition

Given a JD and a pool of CVs, the system returns the top-10 best-matching CVs. The matching signal comes only from keywords extracted from natural-language text; human match labels are never used as input features. The 10 GT pairs are used only for supervised training and evaluation.

1.2 Data

- **10 standard JDs**: the first part of each PDF in `JD + Hired CV examples`; natural-language descriptions.
- **10 hired CVs**: the `CV OF THE HIRED CANDIDATE` part of the same PDFs; one known positive per JD.
- **100 anonymised CVs**: `Anonymised-CVs-100`; unlabelled candidates and unsupervised training data.
- **Candidate pool**: 100 + 10 = **110 CVs**. The 10 hired CVs are not in the 100 anonymised CVs.
- **Excel not used**: the Manatal structured tags are not used in this task.

2. Method

2.1 Local-LLM keyword extraction

We use a locally deployed **Qwen2.5-Coder-14B-Instruct** via LM Studio at temperature 0, with a fixed JSON schema. Each keyword has a term, category and importance (1–3). Results are cached as JSONL with resume support. Across 120 documents the extraction produced 24.5 keywords on average (range 12–46); three very long outputs were retried with a larger token budget and all became valid.

2.2 Keyword merging and clustering

Alias normalisation is followed by local Nomic keyword embeddings (a non-generative model) and cosine agglomerative clustering. Ambiguous terms such as Java/JavaScript and SQL are protected from merging. Each cluster gets a canonical term, yielding a 120-document × vocabulary TF-IDF matrix.

2.3 Autoencoder

The AE is `V → 512 → 256 → 32 → 256 → 512 → V` with GELU, dropout, AdamW and denoising reconstruction. A supervised listwise retrieval loss is added on the bottleneck: the hired CV is the positive and the other 109 CVs are negatives. The AE therefore learns both document representation and matching from the 10 GT pairs. A 16/32/64 bottleneck ablation favoured 32 dimensions.

2.4 Retrieval and evaluation

JDs and CVs are encoded into 32-d unit vectors and ranked by cosine similarity. The primary LOPO protocol holds out one GT pair per fold, trains on the other 9 pairs + 100 unlabelled CVs, and retrieves from the 110-CV pool. We also report an in-sample upper bound and three unsupervised baselines: TF-IDF, keyword overlap and BM25.

3. Results

> **Recall@10 definition:** for each JD, 1 if the hired CV appears in the top 10, else 0; average over 10 queries. With one relevant CV per query, Recall@10 equals HitRate@10.

Method	Recall@10	Recall@5	MRR@10	Mean rank	95% CI (R@10)
TF-IDF cosine	0.800	0.600	0.440	10.6	[0.490, 0.943]
Keyword overlap	0.700	0.500	0.472	12.0	[0.397, 0.892]
BM25	0.800	0.700	0.471	10.0	[0.490, 0.943]
AE + supervised (in-sample)	1.000	1.000	1.000	1.0	[0.722, 1.000]
AE + supervised (LOPO)	0.800	0.400	0.320	9.3	[0.490, 0.943]

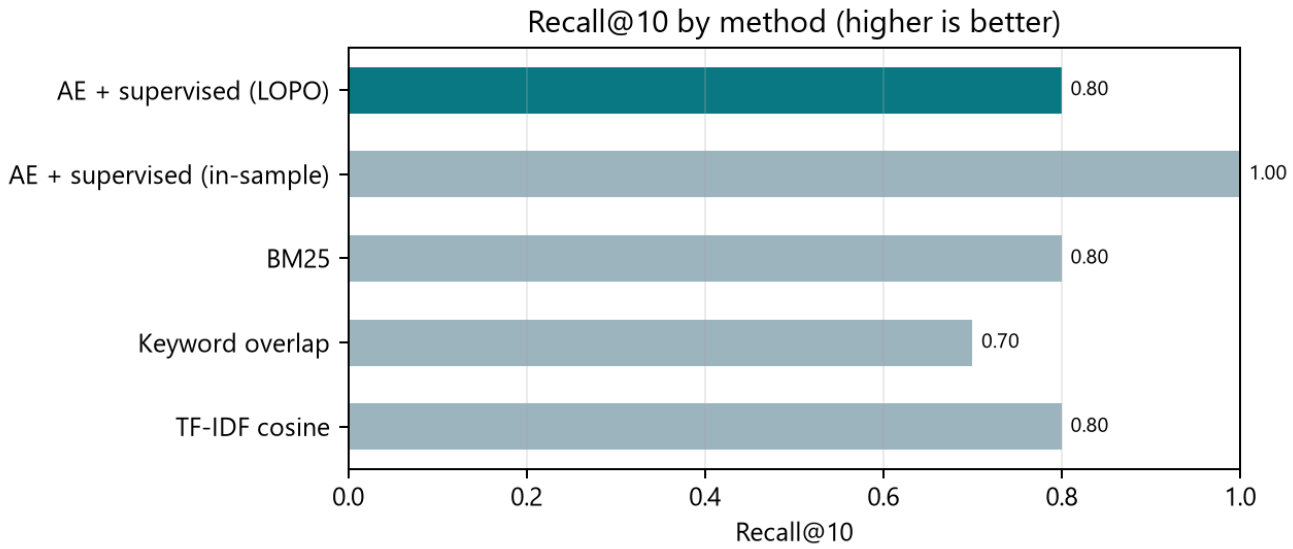


Figure 1. Recall@10 by method.

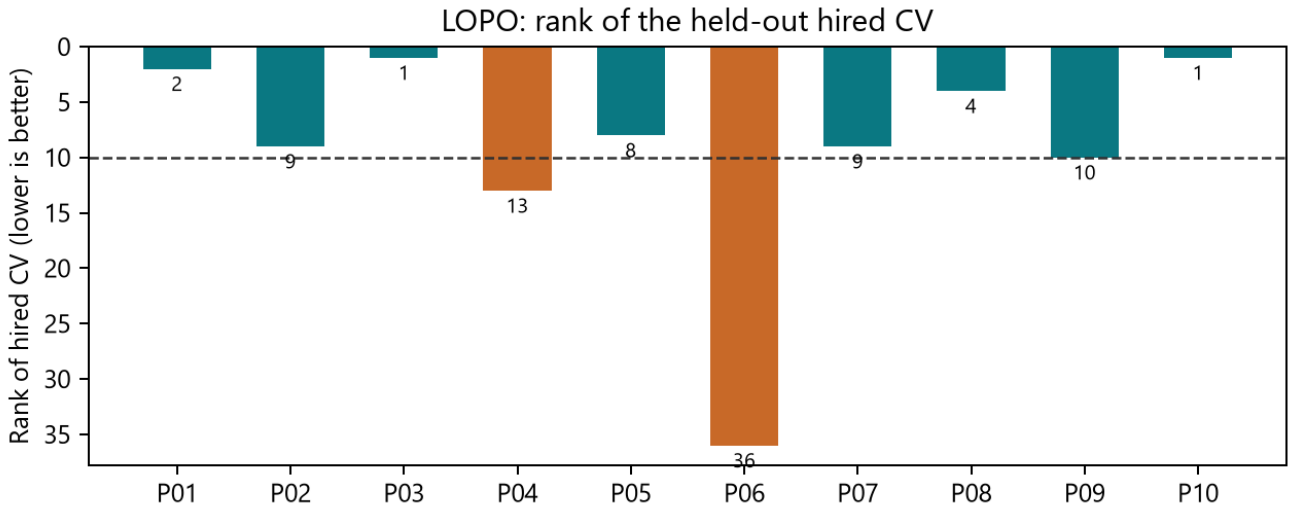


Figure 2. LOPO rank of the held-out hired CV (lower is better).

JD / role	Hired CV	Rank	Hit@10	Top-1 returned
P01 Regional Finance Controller	HIRED_P01	2	✓	CV-082
P02 Sales Manager in Indonesia	HIRED_P02	9	✓	CV-062
P03 GTM Marketing Manager / Lead	HIRED_P03	1	✓	HIRED_P03
P04 Data Engineer	HIRED_P04	13	✗	CV-043
P05 Product Manager — WMS & Warehouse Integrations	HIRED_P05	8	✓	CV-065
P06 Deployment Strategist	HIRED_P06	36	✗	CV-024
P07 Marketing Manager (HK & SG)	HIRED_P07	9	✓	CV-092
P08 Field Service Engineer (FSE)	HIRED_P08	4	✓	CV-050
P09 Tech Lead	HIRED_P09	10	✓	CV-010
P10 Regional Lead Designer UI/UX	HIRED_P10	1	✓	HIRED_P10

Full per-JD top-10 lists are in Appendix C.

4. Error Analysis

The table lists LOPO misses (rank > 10). The last column shows JD keywords missing from the positive CV, separating extraction/clustering errors from genuine must-have gaps.

JD / role	Rank	Top-1	JD∩positive	Missing from positive
P04 Data Engineer	13	CV-043	azure databricks, etl/elt, java, power bi, python, scala	azure synapse, azure, azure data engineer associate, data quality, r, pyspark
P06 Deployment Strategist	36	CV-024		blocker removal, agent refinement, production launches, ai solutions, risk management, ai product

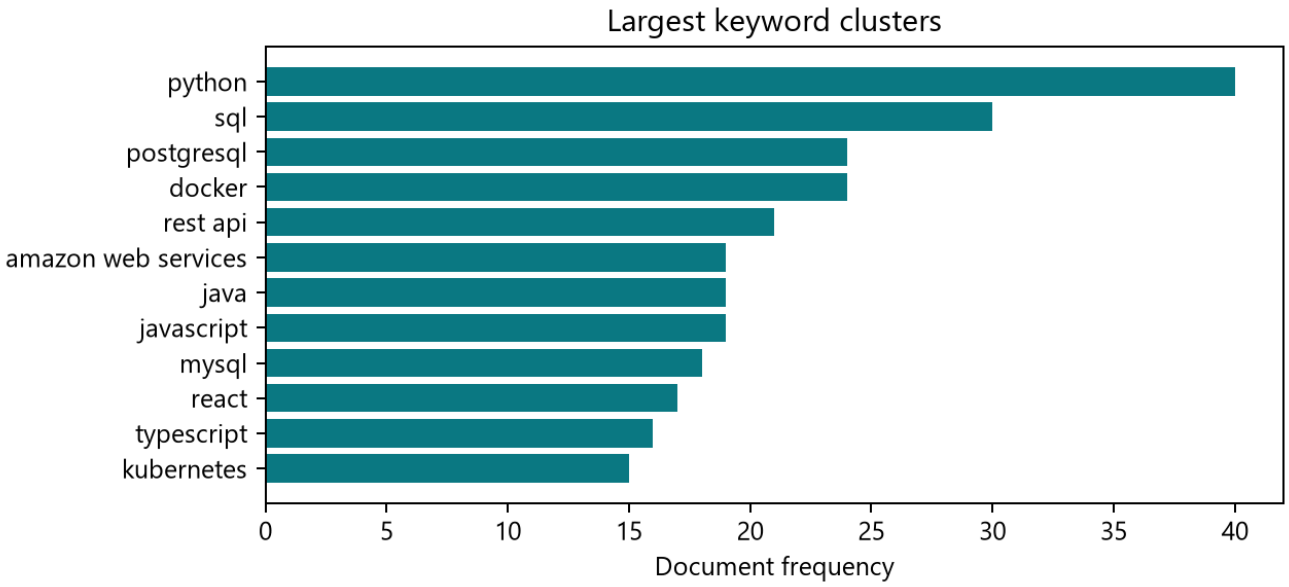


Figure 3. Largest keyword clusters.

4.1 Bottleneck ablation

Bottleneck dim	LOPO Recall@10	Recall@5	MRR@10	Mean rank
16	0.200	0.200	0.100	17.9
32	0.800	0.400	0.320	9.3
64	0.600	0.300	0.266	19.3

16-d underfits, 64-d overfits the 10 pairs; 32-d generalises best.

4.2 Main observations

- **AE matches the baselines on Recall, but has lower MRR:** LOPO Recall@10 = 0.80, the same as TF-IDF / BM25; however AE MRR@10 is 0.32 (TF-IDF 0.44), so it often retrieves the right CV but ranks a similar distractor above it.
- **In-sample is strongly optimistic:** 1.00 after training on all 10 pairs vs 0.80 under LOPO; 10 pairs are easy to memorise, so the report uses LOPO as the primary estimate.
- **Two misses:** P04 (Data Engineer) at rank 13, a near miss with missing Azure Synapse / PySpark terms; P06 (Deployment Strategist) at rank 36, with almost zero keyword overlap because the JD is a project/customer/delivery role while the hired CV is data-science oriented.

- **Hits with low ranks:** P02 (Sales Manager in Indonesia) rank 9, P07 (Marketing Manager (HK & SG)) rank 9, P09 (Tech Lead) rank 10; the positive CV shares few explicit JD keywords and is pulled into the top-10 by the latent vector, while top-1 is often a domain-irrelevant CV (e.g. P08's top-1 is a video-editing CV).
- **Synonyms remain a major error source:** flavour/flavor, key account management/client relationship management, B2B sales/international sales are not fully merged, so P02/P07 positives share only one JD keyword.

5. Improvements

- **Must-have gating:** use JD keywords with importance=3 as a hard gate; down-weight or filter CVs missing critical must-haves.
- **Hybrid retrieval:** combine AE vectors with BM25 / keyword overlap to balance semantics and exact terminology.
- **Better normalisation:** expand the skill synonym table and separate Java/JavaScript, SQL/MySQL, AWS/Azure/GCP.
- **More GT:** 10 pairs are too few; add 30–50 human relevance judgments or run a pooled evaluation.
- **Category-aware encoding:** encode skill/tool/domain blocks separately so generic responsibility words do not dominate hard skills.

6. Limitations

- Only 10 GT pairs; the confidence interval is wide and the query granularity is 0.1.
- Recall treats only the hired CV as relevant; other potentially good CVs are unlabelled, so the metric is conservative.
- The 110-CV pool is small; results should not be extrapolated to large-scale recruiting.
- LLM extraction has category drift and misses; no large-scale human audit was performed.

Appendix

A. LLM prompt

```
Extract 12-25 canonical keywords...
Return JSON: doc_id, doc_type, role_title, seniority,
years_experience, location, languages, keywords[{term, category, importance}].
"languages" must contain ONLY human spoken languages.
Programming languages use category "programming_language".
```

B. Reproducibility

All code and intermediate artefacts are under `HW4_6207/`; run `python src/train_eval.py --mode all`. Random seed is 42.

C. Top-10 retrieval per JD

The table lists the actual top-10 CVs returned for each GT JD under LOPO. ✓ marks the hired CV; if it is absent, its rank is > 10 (P04 Data Engineer = 13, P06 Deployment Strategist = 36).

JD / role	Hired CV rank	Top-10 CVs (1 → 10)
P01 Regional Finance Controller	2	1. CV-082 · **2. HIRED_P01** ✓ · 3. CV-044 · 4. CV-025 · 5. CV-050 · 6. CV-075 · 7. CV-046 · 8. CV-083 · 9. CV-028 · 10. CV-015
P02 Sales Manager in Indonesia	9	1. CV-062 · 2. CV-065 · 3. CV-022 · 4. CV-073 · 5. CV-049 · 6. CV-098 · 7. CV-099 · 8. CV-017 · **9. HIRED_P02** ✓ · 10. CV-083
P03 GTM Marketing Manager / Lead	1	**1. HIRED_P03** ✓ · 2. CV-023 · 3. CV-065 · 4. CV-066 · 5. CV-099 · 6. CV-069 · 7. CV-070 · 8. CV-073 · 9. CV-067 · 10. CV-049
P04 Data Engineer	13	1. CV-043 · 2. CV-031 · 3. CV-083 · 4. CV-022 · 5. CV-037 · 6. CV-041 · 7. CV-087 · 8. CV-040 · 9. CV-035 · 10. CV-061
P05 Product Manager — WMS & Warehouse Integrations	8	1. CV-065 · 2. CV-071 · 3. CV-100 · 4. CV-048 · 5. CV-097 · 6. CV-042 · 7. CV-023 · **8. HIRED_P05** ✓ · 9. CV-070 · 10. CV-022
P06 Deployment Strategist	36	1. CV-024 · 2. CV-016 · 3. CV-085 · 4. CV-010 · 5. CV-023 · 6. CV-066 · 7. CV-019 · 8. CV-009 · 9. CV-058 · 10. CV-018
P07 Marketing Manager (HK & SG)	9	1. CV-092 · 2. CV-099 · 3. CV-065 · 4. CV-023 · 5. CV-073 · 6. CV-022 · 7. CV-083 · 8. CV-035 · **9. HIRED_P07** ✓ · 10. CV-042
P08 Field Service Engineer (FSE)	4	1. CV-050 · 2. CV-063 · 3. CV-083 · **4. HIRED_P08** ✓ · 5. CV-067 · 6. CV-073 · 7. CV-082 · 8. CV-032 · 9. CV-031 · 10. CV-033
P09 Tech Lead	10	1. CV-010 · 2. CV-035 · 3. CV-037 · 4. CV-041 · 5. CV-083 · 6. CV-040 · 7. CV-092 · 8. CV-067 · 9. CV-042 · **10. HIRED_P09** ✓
P10 Regional Lead Designer UI/UX	1	**1. HIRED_P10** ✓ · 2. CV-037 · 3. CV-070 · 4. CV-042 · 5. CV-035 · 6. CV-041 · 7. CV-061 · 8. CV-040 · 9. CV-022 · 10. CV-074